# MZUZU UNIVERSITY

## FACULTY OF SCIENCE, TECHNOLOGY AND INNOVATION

### DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

### BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY

## BICT 1102: INTRODUCTION TO PROGRAMMING WITH C

### END OF SEMESTER EXAMINATION

**DATE: JULY 2023** **TIME ALLOWED: 3 HOURS**

**INSTRUCTIONS**

This paper contains **TWO** pages. **PLEASE CHECK**

There are **FOUR** questions in this paper. **ANSWER ALL.**

Marks are indicated for each question.

**Question One**

a) Explain the use of #include directive. **[3 marks]**

b) Describe the difference between the following in C Programming:

i. Logical AND (&&) and Bitwise AND (&) operators. **[6 marks]**

ii. Array of pointer and Pointer to an array. **[6 marks]**

c) Write a c program to display the following palindrome.

**0 0 1**
**0 1 0**
**1 0 0** **[10 marks]**

**Question Two**

a) Using **one** example, explain the use of the GOTO statement. **[3 marks]**

b) Write a program to initialize three different strings and perform the following functions:

i. Copy STRING1 into STRING3. **[5 marks]**

ii. Concatenate STRING1 and STRING2 **[5 marks]**

c) Explain the main difference between pass-by-value and pass-by-reference in function parameter passing. **[6 marks]**

d) Write a C program that calculates the factorial of a given positive integer using a recursive function. The program should prompt the user to enter a positive integer and display its factorial. **[6 marks]**

**Question Three**

a) As part of a system to store the inventory of parts for a robot, you have been tasked with creating a data structure for the storage and retrieval of electronic component data. Using a C struct, show how you would accomplish the following:

i. Declare a structure for representing electronic components:
*component type, sub-type*e.g. capacitor, electrolytic (char[20], char[20])
*value, exponent* e.g. 47, -6 would represent 47 microFarad (float, int) *voltage rating* e.g. 200 (volts assumed) (float). **[6 marks]**

ii. Use typedef to define a new Component data type. **[4 marks]**

iii. Declare a pointer to a Component and show how to access the value of the component using the pointer. **[4 marks]**

iv. Write a function prototype for a function called ***select_component_value*** which has the following parameters: an array of Components, a type (e.g. Capacitor, Resistor), a specified value, a minimum voltage rating. The function should return a pointer to a single Component. **[5 marks]**

**b)** Briefly describe the difference between a Union and a Struct in C programming. **[6 marks]**

**Question Four**

a) Describe what will be the output for the following code snippet: **[5 marks]**

```
#include <stdio.h>
int main() {
   int arr[] = {1, 2, 3, 4, 5};
   int *ptr = arr;
   printf("%d\n", *(ptr + 2));
   return 0;
}
```

b) Consider a system development project that allows users to browse products, add items to their cart, and check out. Write a C programme that accomplishes the following:

- Calculate the total cost of the items in the cart, apply applicable discounts or taxes, and display the final amount to the user.
- Prompt users to enter the quantity and price of each item they wish to purchase.
- Calculate and display the total cost, including any discounts or taxes applied.

**[20 marks]**

**END OF QUESTIONS**